

COMP1003 Computer Organization

Assignment 1

Q1

$$f(x, y, z) = x \cdot (y + \overline{z + x}) + \overline{z} \cdot \overline{y} \cdot (x + z)$$

(a) Truth table

Each row is obtained by evaluating the two products and then OR-ing them. Term 1 is $x \cdot (y + \overline{z + x})$ and term 2 is $\overline{z} \cdot \overline{y} \cdot (x + z)$.

x	y	z	$\overline{z + x}$	$y + \overline{z + x}$	term 1	$\overline{z} \cdot \overline{y}$	$x + z$	term 2	f
0	0	0	1	1	0	1	0	0	0
0	0	1	0	0	0	1	1	1	1
0	1	0	1	1	0	1	0	0	0
0	1	1	0	1	0	0	1	0	0
1	0	0	0	0	0	1	1	1	1
1	0	1	0	0	0	1	1	1	1
1	1	0	0	1	1	1	1	1	1
1	1	1	0	1	1	0	1	0	1

$f = 1$ on minterms m_1, m_4, m_5, m_6, m_7 .

(b) Disjunctive normal form

Reading the rows where $f = 1$:

$$\begin{aligned} f(x, y, z) &= m_1 + m_4 + m_5 + m_6 + m_7 \\ &= \overline{x}\overline{y}z + x\overline{y}\overline{z} + x\overline{y}z + xy\overline{z} + xyz. \end{aligned}$$

(c) Karnaugh map

Columns are in Gray-code order, so cells that share an edge differ in exactly one variable.

		yz			
		00	01	11	10
x	0	0	1	0	0
	1	1	1	1	1

- Red quad, the row $x = 1$: every cell is 1, so this group is x .

- Blue pair, the column $yz = 01$: these two cells are $\bar{y}z$. The cell $x\bar{y}z$ belongs to both groups, which is allowed.

No 1 is left outside these groups, so

$$f(x, y, z) = x + \bar{y}z.$$

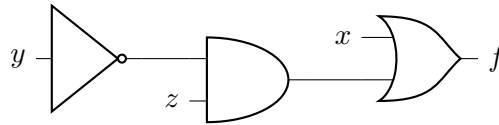
The same expression follows from the original formula. By De Morgan, $\overline{z + x} = \bar{z}\bar{x}$ and $\overline{zy} = \bar{z} + \bar{y}$, hence

$$\begin{aligned} f &= x(y + \bar{x}\bar{z}) + (\bar{y} + \bar{z})(x + z) \\ &= xy + (\bar{y}x + \bar{y}z + x\bar{z}) \\ &= x(y + \bar{y}) + \bar{y}z + x\bar{z} \\ &= x + \bar{y}z, \end{aligned}$$

where $x + x\bar{z} = x$.

(d) Logic circuit of the minimized function

$f = x + \bar{y}z$ uses one NOT, one AND, and one OR.



Q2

(a) 152_{dec} to binary

Divide by 2 and record the remainders. The binary digits are those remainders read from last to first.

$$\begin{array}{ll} 152 = 2 \cdot 76 + 0 & \text{remainder 0} \\ 76 = 2 \cdot 38 + 0 & \text{remainder 0} \\ 38 = 2 \cdot 19 + 0 & \text{remainder 0} \\ 19 = 2 \cdot 9 + 1 & \text{remainder 1} \\ 9 = 2 \cdot 4 + 1 & \text{remainder 1} \\ 4 = 2 \cdot 2 + 0 & \text{remainder 0} \\ 2 = 2 \cdot 1 + 0 & \text{remainder 0} \\ 1 = 2 \cdot 0 + 1 & \text{remainder 1} \end{array}$$

$$152_{\text{dec}} = 10011000_2.$$

Check: $2^7 + 2^4 + 2^3 = 128 + 16 + 8 = 152$.

(b) Unsigned 11001010_{bin} to decimal

$$\begin{aligned} 11001010_2 &= 2^7 + 2^6 + 2^3 + 2^1 \\ &= 128 + 64 + 8 + 2 \\ &= 202_{\text{dec}}. \end{aligned}$$

(c) Signed 11001010_{sig} to decimal

The subscript sig is read as sign-magnitude. Part (d) asks for two's complement separately, so the sign bit here is not a two's-complement sign.

The leading bit is 1, so the number is negative. The magnitude is the remaining seven bits:

$$1001010_2 = 2^6 + 2^3 + 2^1 = 64 + 8 + 2 = 74.$$

Therefore

$$11001010_{\text{sig}} = -74_{\text{dec}}.$$

(The same eight bits, if they were instead an 8-bit two's-complement integer, would equal -54 . That reading is not used here.)

(d) -152_{dec} as 8-bit two's complement

An n -bit two's-complement integer lies in $[-2^{n-1}, 2^{n-1}-1]$. For $n = 8$ the interval is $[-128, 127]$. Since $-152 < -128$, -152 has no 8-bit two's-complement representation.

The shortest width that contains it is 9 bits, whose range is $[-256, 255]$. Start from the positive value and apply invert-and-add-one:

$$\begin{aligned} +152 &= 010011000_2, \\ \text{invert} &= 101100111_2, \\ +1 &= 101101000_2. \end{aligned}$$

So the 9-bit two's-complement form is 101101000_2 . Inverting that pattern and adding one recovers $010011000_2 = 152$, which confirms the sign.

(e) $FAC E_{\text{hex}}$ to octal

Replace each hex digit by four bits, then regroup by three bits from the right.

$$\begin{aligned} F &= 1111, & A &= 1010, & C &= 1100, & E &= 1110, \\ FAC E_{16} &= 1111101011001110_2. \\ 1111101011001110_2 &= 175316_8. \end{aligned}$$

Decimal check: $15 \cdot 16^3 + 10 \cdot 16^2 + 12 \cdot 16 + 14 = 64206$, and $64206_{10} = 175316_8$.

(f) Half-precision 1011001110001100 to decimal

IEEE 754 binary16 has 1 sign bit, 5 exponent bits (bias 15), and 10 fraction bits:

$$1 \ 01100 \ 1110001100.$$

- Sign bit 1: the value is negative.
- Stored exponent $01100_2 = 12$. It is neither 00000 nor 11111 , so the number is normal. The true exponent is $12 - 15 = -3$, and the leading significand bit is 1.
- Fraction $1110001100_2 = 908$, so

$$1 + \frac{908}{1024} = 1 + \frac{227}{256} = \frac{483}{256}.$$

$$\begin{aligned} 1011001110001100 &= -\frac{483}{256} \cdot 2^{-3} \\ &= -\frac{483}{2048} \\ &= -0.23583984375_{\text{dec}}. \end{aligned}$$

Q3

Unsigned integers, with no fixed word size:

$$1111001_2 \times 10011_2 - 1010110_2.$$

(a) Binary value

The multiplier bits, from the right, are 1, 1, 0, 0, 1. The two zero partial products are omitted; the three nonzero ones are shifted by 0, 1, and 4 places.

$$\begin{array}{r} 1111001 \\ \times \quad 10011 \\ \hline 1111001 \\ 1111001 \\ 1111001 \\ \hline 10001111011 \end{array}$$

The first two partial products add as

$$1111001_2 + 11110010_2 = 101101011_2,$$

and

$$101101011_2 + 11110010000_2 = 10001111011_2.$$

Subtract the third operand. Align 1010110_2 on the right:

$$\begin{array}{r} 10001111011 \\ - 00001010110 \\ \hline 100010100101 \end{array}$$

The only borrow starts at bit 2 (value bits numbered from 0 on the right): that bit is $0 - 1$, so it borrows from bit 3 and the result bit is 1. No further borrow is generated. The remaining bits subtract directly and give 100010100101_2 .

$$1111001_2 \times 10011_2 - 1010110_2 = 100010100101_2.$$

(b) Decimal check

$$1111001_2 = 64 + 32 + 16 + 8 + 1 = 121,$$

$$10011_2 = 16 + 2 + 1 = 19,$$

$$1010110_2 = 64 + 16 + 4 + 2 = 86.$$

$$121 \times 19 - 86 = 2299 - 86 = 2213.$$

$$2213 = 2^{11} + 2^7 + 2^5 + 2^2 + 2^0 = 100010100101_2,$$

which matches part (a).